



Program	BACHELOR OF VOCATION (B.Voc.)	Semester - 4
Type of Course	-	
Prerequisite		
Rationale	-	
Effective From A.Y.	2025-26	

Teaching Scheme (Contact Hours)				Examination Scheme				
Lecture	Tutorial	Lab	Credit	Theory Marks		Practical Marks		Total Marks
				SEE T	IAT	SEE P	CCE	
0	0	2	1	-	-	30	20	50

SEE - Semester End Examination, IAT - Internal Assessment Test, CCE - Continues & Comprehensive Evaluation

Course Content		T - Teaching Hours W - Weightage	
Sr.	Topics	T	W
1	Practical 1 1. Write a program to access a class from another package 2. Demonstrate the use of built-in packages (java.util, java.io, etc.)	2	5
2	Practical 2 Create a package containing multiple classes and access them	2	5
3	Practical 3 1. Write a program to create a new file using the File class 2. Write a program to write data into a file using FileWriter	2	5
4	Practical 4 Write a program to read data from a file using FileReader/BufferedReader	4	5
5	Practical 5 Implement TCP Server for transferring files using Socket and ServerSocket	4	20
6	Practical 6 Implement student registration form with enrollment number, first name, last name, semester, contact number. Store the details in the database. Also implement search, delete and modify facilities for student records	4	20
7	Practical 7 Write a Servlet program to print system date and time	4	10
8	Practical 8 Create a Servlet file and study web descriptor files	4	10
9	Practical 9	2	10



Course Content		T - Teaching Hours W - Weightage	
Sr.	Topics	T	W
	Write a program that shows the usage of JSP Scriptlets, Expressions & Declarations		
10	Practical 10 Write a program that shows Reading Form Data in JSP	2	10
Total		30	100

Course Outcomes	
At the end of this course, students will be able to:	
C01	To Understand the purpose of packages in Java
C02	To Understand the concept of java networking
C03	To Implement Database connectivity in Java for given applications
C04	To Implement a webpage with dynamic content and server side web application using Servlet
C05	To Use JSP elements effectively to build dynamic web pages

Reference Books	
1.	Black Book JAVA Server Programming J2EE By Kathy Walrath Dream Tech Publishers 1st
2.	Complete Reference J2EE By James Keogh McGraw Publication
3.	Professional JAVA Server Programming By Subrahmanyam Allamaraju, Cedric Buest Wiley Publication
4.	SCWCD By Matthew Scarpino, Hanumant Deshmukh, Jignesh Malavie Manning Publication
5.	Core Java, Volume II: Advanced Features By Cay Horstmann and Gary Cornell Pearson Publication
6.	JAVA Persistence with Hibernate By Christian Bauer, Gavin King
7.	Spring in Action By Craig walls Manning Publication 3rd
8.	Hibernate By Jeff Linwood and Dave Minter, Beginning Après publication 2nd
9.	Java Server Faces in Action By Kito D. Mann Manning Publication
10.	JDBC API Tutorial and Reference By Maydene Fisher, Jon Ellis, Jonathan Bruce Addison Wesley 3rd
11.	Beginning JSP, JSF and Tomcat By Giulio Zambon Apress
12.	JSF 2.0 CookBook By Anghel Leonard PACKT publication



List of Practical

1.	Practical 1
2.	Practical 2
3.	Practical 3
4.	Practical 4
5.	Practical 5
6.	Practical 6
7.	Practical 7
8.	Practical 8
9.	Practical 9
10.	Practical 10